

Seven gene sets. One auditable curation.

A concise record of the publication rules, pathway selection, identity checks, and overlap that produced the stocker-ready inputs.

7

stocker CSVs

2

analysis families

4

primary sets

3

GO term sets

Two curation families—analyzed separately

Primary evidence sets and CSW-derived GO term-grouped sets use separate overlap analyses.

FAMILY 1 • PRIMARY CURATION SETS

01

6

genes

Mechanistic

Consistent/correlated + published effect supporting homeostatic function

02

20

genes

Homeostatic genes

Opposing history ↔ rebound correlations: 15 + 5

03

97

genes

CSW 4+ genes

Consistent in ≥4 of 7 datasets

04

4

genes

HLH genes

Potential upstream regulators

FAMILY 2 • CSW-DERIVED GO TERM-GROUPED SETS (FC0.5 / FC1)

05

99

genes

Ribosomal

DAVID term hits; approved keywords

06

167

genes

Mitochondrial

DAVID mito/metabolism term hits

07

5

genes

Immune

DAVID immune / Toll / Imd term hits

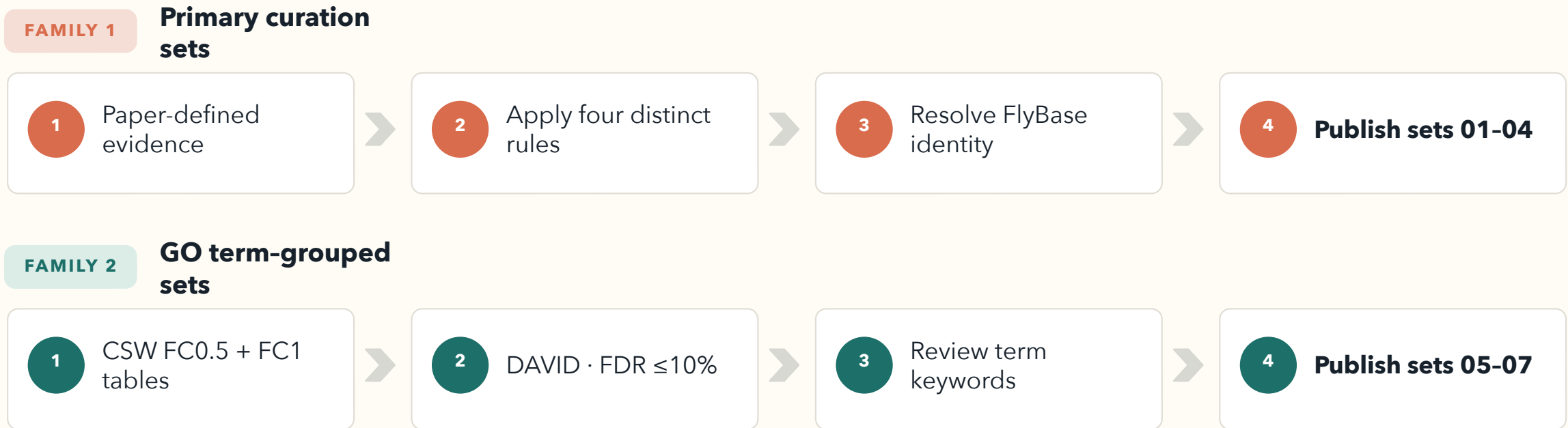
RULE

Keep them separate

Report overlap within each family—never as one seven-set comparison.

Two curation tracks converge on one controlled folder

They share identity gates and file standards—but their overlap analyses remain separate.

**902**

source observations

791

unique source FBgns

 ≥ 2

minimum pathway frequency

7

experiments

Thresholds remain separate: FC0.5 and FC1 are never collapsed into a max-frequency field.

Mechanistic and Homeostatic are different evidence categories

The six-gene functional set is distinct from the 20-gene opposing history-rebound correlation set.

SET 01 · 6 GENES

Mechanistic

Consistent or correlated sleep/wake genes with published effects in a direction consistent with potential homeostatic function.

- Expression is consistent across conditions or correlated with sleep/wake history.
- Published perturbation affects sleep/wake in a direction compatible with negative feedback.
- The paper highlights six such genes.

unc79 · SIFa · rumpel · AstA-R2 · Trhn · RpL23

SET 02 · 20 GENES

Homeostatic genes

15

History+ / Rebound–

5

History– / Rebound+

Genes correlate with sleep/wake history before sampling and with rebound in the opposite direction afterward.

Identity correction

Source row trh / FBgn0262139 is resolved from publication context to Trhn / FBgn0035187 (tryptophan hydroxylase). The wrong FBgn is never retained.

CSW consistency and HLH regulation are separate categories

Set 03 measures recurrence across seven datasets; Set 04 identifies potential upstream regulators.

SET 03 · 97 GENES

CSW 4+ genes

97

Current output · FC0.5 wake
frequency 4-6

7

also qualify at FC1

0

sleep genes at
frequency 4

Category rule

- Consistent sleep/wake genes evident across four or more of seven datasets.
- The current output contains 97 FC0.5 wake genes; seven also qualify at FC1.
- No sleep-direction gene reaches frequency 4 in this run.

SET 04 · 4 GENES

HLH genes

Named in Results / Table S1 after
HOMER identified a CAGCTG E-
box upstream of wake-induced
genes.

bigmax FBgn0039509

HLH3B FBgn0011276

E(spl)m3-HLH FBgn0002609

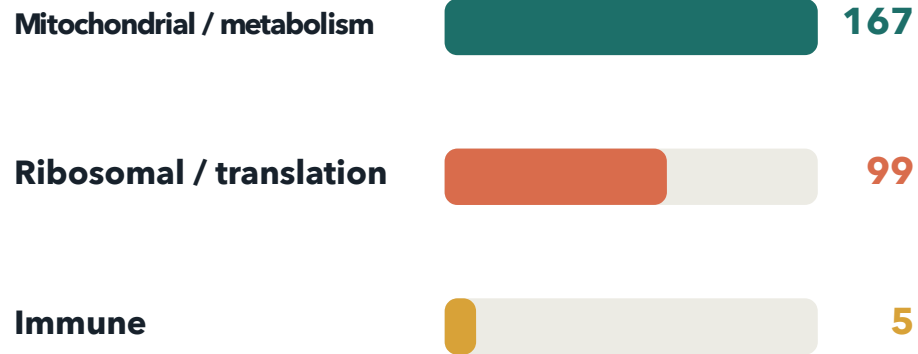
E(spl)mbeta-HLH FBgn0002733

CSW-derived GO term-grouped genes retain threshold provenance

Each gene records its source as FC0.5, FC1, or both; only hits from approved FDR-passing terms enter sets 05-07.

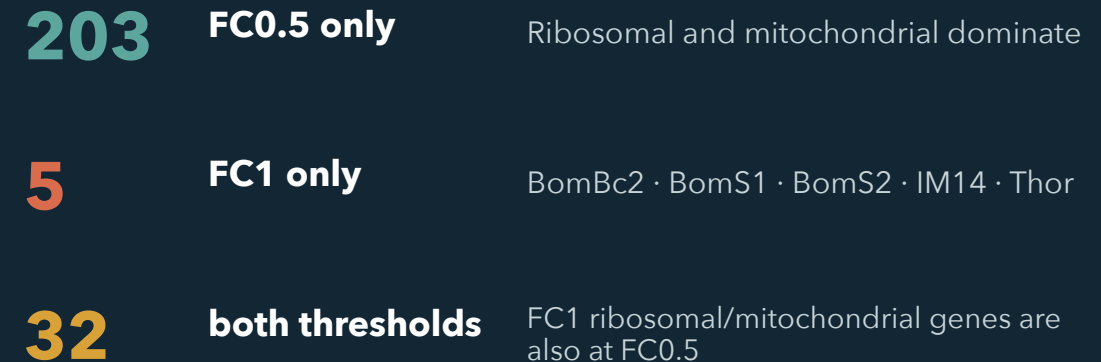


GO term-grouped outputs



All three outputs are wake-direction genes.

Threshold provenance · 240 unique FBgns



Key takeaway Only the five immune genes are FC1-exclusive.

Identity resolution is conservative by design

Current symbols can pass directly; ambiguous synonyms require biological context or a case-supported source match.

PASS**Trhn****FBgn0035187**

Current symbol · tryptophan hydroxylase neuronal · publication-resolved in Homeostatic.

REJECT**trh****FBgn0262139**

Current symbol · trachealess · must not enter any generated identity field for this publication row.

REVIEW**Trh****ambiguous**

Synonym of Hn, Trhn, and trh · never auto-approved without biological context.

Approval gates

- Exact current symbol
- Publication-supported correction
- Unique case-insensitive current-symbol match
- Otherwise: stop for review

Membership key

flybase_gene_id

Symbols are display labels. All overlap, deduplication, and set membership use FBgn only.

Primary-set overlap is limited to three genes

Mechanistic, Homeostatic genes, CSW 4+ genes, and HLH genes are compared only with one another.

6

Mechanistic

20

Homeostatic genes

97

CSW 4+ genes

4

HLH genes

Observed shared genes

Mechanistic × Homeostatic genes**2**

Trhn · AstA-R2

Mechanistic × CSW 4+ genes**1**

RpL23

All other pairings**0**

No shared FBgns

Within Family 1

124

unique FBgns

121

in exactly one set

3

in exactly two sets

0

in three or four sets

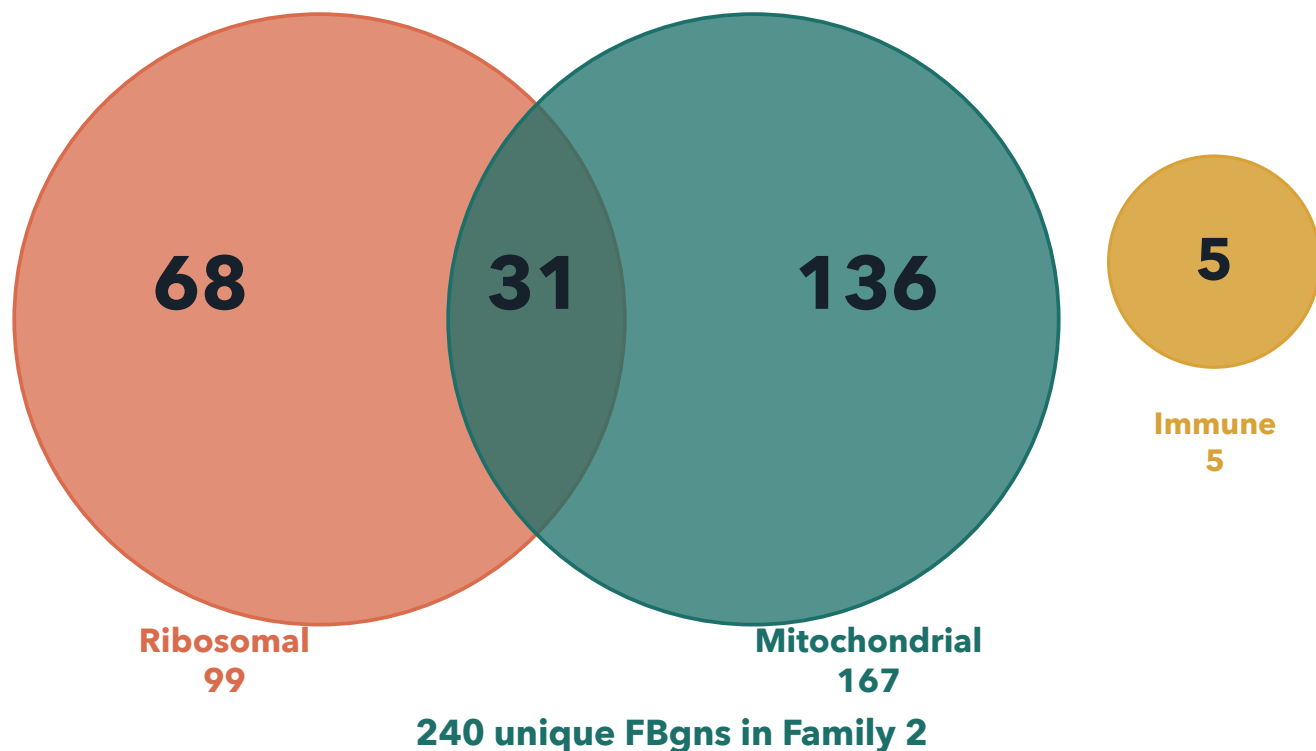
HLH is fully disjoint from the other primary sets.

GO term-grouped overlap is confined to ribosomal × mitochondrial

The three pathway buckets are compared only within Family 2; threshold provenance remains a separate attribute.

GO term-group membership

Schematic · counts shown



What the overlap means

31 genes occur in both ribosomal and mitochondrial buckets

5 immune genes are fully disjoint

0 immune overlaps with either other GO bucket

The 31-gene overlap is intentional: mitochondrial translation terms match both keyword groups.

The contract is simple—and strict.

7

Exactly seven CSVs
in the discovery
folder

2

Required keys:
ext_gene and
flybase_gene_id

0

No overlap, GO, or
audit CSVs beside
the inputs

RUN

```
python -m fl_ai_reagent_stocker run ./data/gene_sets/Tx-Omics_Revision/ \
--config ./data/config/stock_split_config_priority_example.json
```

Seven inputs

- 01 Mechanistic
- 02 Homeostatic genes
- 03 CSW 4+ genes
- 04 HLH genes
- 05 Ribosomal
- 06 Mitochondrial
- 07 Immune

Stability rule

Discovery is recursive: every extra *.csv changes the run. Keep analysis outputs elsewhere.